



NETROKONA UNIVERSITY
 Department of Computer Science and Engineering
Netrokona University Computer Club (NEUCC)



NEUCC — Web Application

Complete Software Requirement Specification

Public Website • Role-Based User Panel • Course: Software Engineering Lab

Field	Details
Project Title	Netrokona University Computer Club — Web Application
Document Type	Complete Software Requirement Specification (Public Website + Role-Based User Panel)
Submitted To	Abdullah Al Siam (Supervisor)
Date	27 August 2026

Team Members

Name	Registration No.	Exam Roll
Najmul Huda	202204012	412
Zuel Rana	202204011	411
Md. Samiul Islam Pias	202204024	423

Table of Contents

① Introduction	4
1.1 Purpose.....	4
1.2 Project Overview	4
1.3 Scope	4
1.3.1 Public Website (14 pages).....	4
1.3.2 Internal User Panel (4 roles)	4
② Objectives	5
③ Public Website — Page-wise Functional Requirements.....	5
3.1 Home Page.....	5
3.2 Event Tab (Event Listing Page).....	6
3.3 About Page.....	6
3.4 Executives Page	7
3.5 Contests Page	7
3.6 Gallery Page.....	7
3.7 Achievements / Awards Page	7
3.8 Announcements Page.....	8
3.9 Sponsors / Partners Page.....	8
3.10 Join Us Page.....	8
3.11 Contact Page.....	8
3.12 Privacy Policy Page.....	9
3.13 Terms & Conditions Page.....	9
3.14 Footer	9
④ Governance Model (from the Club Constitution)	10
4.1 Membership (Constitution Ch. 2)	10
4.2 Club Structure (Constitution Ch. 3).....	10
4.3 Roles & Responsibilities (Constitution Ch. 4, selected)	10
4.4 Assembly & Elections (Constitution Ch. 5–6).....	11
4.5 Finance & Policy (Constitution Ch. 8–9).....	11
⑤ User Panel — Roles & Committee Lifecycle.....	11
5.1 User Roles.....	11
5.2 Committee Lifecycle & Governance Workflow.....	12
5.3 System Requirements for the Lifecycle	13
5.4 Role-wise Permission Matrix	13
⑥ User Panel — Module-wise Functional Requirements.....	14
6.1 Election Module.....	14
6.1.1 Candidate Nomination, Fee & Symbol Allotment Workflow	15
6.2 Event Management Module.....	16
6.3 Notice, Resolution & Document Module.....	16
6.3.1 Notice Pad (Letterhead) Template.....	17
6.3.2 Notice Pad Generator — Workflow.....	18
6.3.3 Agentic AI Content Generator.....	18
6.3.4 Notice Visibility & Audience Scope.....	19
6.4 Attendance Module.....	20

6.5 Dashboard.....	20
6.6 Treasury / Finance Module.....	21
⑦ Non-Functional Requirements.....	22
⑧ Technology Stack.....	23
⑨ Site Navigation Flow.....	23
⑩ Reference Documents.....	24
⑪ Conclusion.....	24

① Introduction

1.1 Purpose

This document describes the complete requirements for developing a production-ready, full-stack web application for the Netrokona University Computer Club (NEUCC). The application has two halves that must work together as one system:

- A public-facing website that represents the club's identity online, showcases its activities, events, achievements, and members, and lets students, faculty, and visitors learn about and connect with the club.
- A role-based internal user panel that digitizes the club's governance exactly as defined in its Constitution (Amended 2026) — membership, committees, elections, event management, notices/resolutions, and meeting attendance — enforced through login-based roles and permissions.

1.2 Project Overview

The NEUCC Web Application will be the digital home of the club. Visitors will see the club's mission, events, achievements, executive committee, and sponsors, and will have multiple channels to get in touch or join. Behind the login, the club's actual governance — committee formation, elections, event control, notices, and attendance — will be run through the same application, driven by the rules already established in the Constitution and the club's Monthly Meeting Resolutions.

1.3 Scope

1.3.1 Public Website (14 pages)

- Home Page
- Event Tab (Event Listing Page — includes detailed event view)
- About Page
- Executives Page
- Contests Page
- Gallery Page
- Achievements / Awards Page
- Announcements Page
- Sponsors / Partners Page
- Join Us Page
- Contact Page
- Privacy Policy Page
- Terms & Conditions Page
- Footer (shared component across all pages)

1.3.2 Internal User Panel (4 roles)

- Advisory Board (উপদেষ্টা মন্ডলী)
- Election Committee (নির্বাচন কমিটি)
- Executive Committee (এক্সিকিউটিভ কমিটি)
- Club Member (ক্লাব মেম্বর)

2 Objectives

- Create a professional online identity for the University Computer Club.
- Provide up-to-date information about club events, workshops, and seminars.
- Allow visitors to view detailed information about each event, including guests and speakers.
- Share the club's mission, vision, and history with students and faculty.
- Showcase the club's achievements, awards, and recognition.
- Highlight sponsors and partner organizations.
- Provide a simple and reliable way for people to contact and join the club.
- Maintain transparency through clear privacy and usage policies.
- Build a responsive, mobile-friendly, easy-to-navigate website with dark/light theme support.
- Digitize the club's constitutional governance: committee formation, one-year executive terms, elections, and accountable record-keeping.
- Enforce strict role-based access so each committee post can only perform the actions the Constitution assigns to it.

3 Public Website — Page-wise Functional Requirements

3.1 Home Page

The landing page of the website. It gives visitors a first impression of the club and directs them to other sections.

Requirement	Description
Hero Section	Club name, logo, tagline, and a short welcome message with a call-to-action button (e.g., "Join Us" / "View Events").
About Snapshot	A brief 2–3 line introduction of the club with a "Read More" link to the About page.
Club in Numbers	Animated stat counters — Total Members, Events Per Year, Workshops Conducted, Years of Excellence.
Chairperson's Message	Photo, name, designation, and a short quote/message from the Department Chairperson.
Moderator's Message	Photo, name, designation, and quote from faculty moderator(s) — displayed as a card grid if multiple.
Upcoming Events Preview	Cards showing the next 2–3 upcoming events with a "View All Events" link to the Event Tab.
Highlights / Achievements	Showcase of notable achievements — Programming contest rank, CTF wins, ICPC rank, IUPC rank, hackathon results, or workshops conducted.
Featured Collaborations	Scrollable strip of partner clubs/universities with logos and short descriptions.
Gallery Preview	A few images from past events for visual appeal, linking to the full Gallery page.
Call-to-Action Banner	Membership/registration invitation section linking to the Join Us page.
Navigation Bar	Links to Home, Events, About, Executives, Contests, Gallery, Achievements, Announcements, Sponsors, Contact, and Join Us.

3.2 Event Tab (Event Listing Page)

Displays all events organized or planned by the club in a structured, browsable list. Full event details are shown within this page instead of a separate details page.

Requirement	Description
Event Cards/List	Each event shown with title, date, time, venue, and a short description.
Category/Filter	Filter events by type (Workshop, Seminar, Competition, Meetup) and by status (Upcoming / Past).
Search Bar	Allow visitors to search events by name or keyword.
Expandable Card / Modal	Clicking a card expands or opens a modal showing full description, agenda, and speaker/guest information.
Guest List	Chief Guest, Special Guest(s), and Session Chair, listed with designation.
Participant Count	Total number of participants shown for past events.
Registration Link	Direct registration link or button for upcoming events.
Social Share	Option to share an event on social media.
Pagination	Load events in pages if the list becomes long.

3.3 About Page

Introduces the club's background, mission, and vision.

Requirement	Description
Club Mission & Vision	A clear statement of what the club stands for and aims to achieve.
History	When and why the club was founded, and key milestones.
Faculty Advisor(s)	Name and message from the faculty moderator(s).
Achievements Summary	Brief highlight list linking to the full Achievements page.
Constitution Access	A public link/download button for the full Club Constitution PDF, so the Advisory Board and general Club Members — who do not have login accounts — can read it without needing to sign in.

3.4 Executives Page

Introduces the current executive committee running the club.

Requirement	Description
Committee Grid	Photos, names, and designations of all committee members (President, VP, Secretary, and other posts).
Position Hierarchy	Members arranged by rank/position, from top leadership down.
Social Links	LinkedIn/GitHub/Facebook links for each executive (optional).
Past Committees (optional)	Archive of previous years' committees for continuity.

3.5 Contests Page

Lists programming contests, CTFs, and hackathons organized or attended by the club.

Requirement	Description
Contest List	Name, date, type (Programming / CTF / Hackathon), and result.
Leaderboard/Winners	Names/teams of winners with rank.
Registration Link	Link to register for ongoing or upcoming contests.

3.6 Gallery Page

A complete photo and video archive of club events.

Requirement	Description
Photo/Video Grid	All event media displayed as a thumbnail grid.
Filter by Event/Year	Filter media by specific event or year.
Lightbox View	Clicking a thumbnail opens a larger pop-up view.

3.7 Achievements / Awards Page

Showcases the club's recognitions and accomplishments over time.

Requirement	Description
Achievement Timeline	Chronological listing of awards and recognitions received by the club.
Award Details	Short description, date, and awarding organization for each achievement.
Certificate/Photo Proof	Thumbnail of certificate or award ceremony photo.

3.8 Announcements Page

A news-feed style page for club notices and updates. Content here is populated from the internal Notice/Resolution Module (see Section 6.3), published by the Notice & Publication Team — Information Secretary, Assistant Information Secretary, Editor Secretary, Assistant Editor Secretary, Publicity & Publication Secretary, and Assistant Publicity & Publication Secretary. Only notices tagged General/Public scope (Section 6.3.4) appear here; Internal and Election-scope notices are never shown on this public page.

Requirement	Description
Announcement Feed	Recent announcements/updates listed by date, newsfeed style.
Pin Important Notices	Ability to pin urgent notices to the top of the feed.
Category Tag	Tags such as Result, Notice, or Event Update to distinguish announcement types.

3.9 Sponsors / Partners Page

Recognizes organizations that support the club.

Requirement	Description
Sponsor Logos	Logos grouped by tier (Platinum / Gold / Silver).
Partnership Description	Short description of the relationship with each sponsor.
Become a Sponsor CTA	Contact link/form for prospective sponsors.

3.10 Join Us Page

A dedicated page for prospective members to apply. Submissions here feed into the Membership records used by the internal panel (Constitution §2).

Requirement	Description
Membership Form	Name, Student ID, Batch, Email, and area of interest.
Benefits Section	Summary of benefits of becoming a member.
FAQ	Common questions and answers about membership.

3.11 Contact Page

Provides ways for visitors to reach out to the club.

Requirement	Description
Contact Form	Name, email, subject, and message fields with a submit button.
Direct Contact Info	Official email address and phone number (if applicable).
Social Media Links	Facebook, GitHub, Discord, LinkedIn, etc.
Location Map	Embedded map showing the university/department location.
Office Hours	When the club is available to meet in person (optional).

3.12 Privacy Policy Page

Requirement	Description
Information Collected	What data is collected (name, email, messages, cookies, etc.).
Purpose of Use	Why the data is collected and how it will be used.
Data Protection	Statement on how submitted data is kept secure.
Third-Party Sharing	Whether data is shared with any third party (usually stated as "No").
Contact for Privacy Concerns	An email/contact for privacy-related queries.

3.13 Terms & Conditions Page

Requirement	Description
Acceptable Use	Rules for how visitors may use the website content.

Requirement	Description
Membership Rules	Basic conditions for joining or participating in club events.
Content Ownership	Statement on ownership of the club's logo, content, and materials.
Liability Disclaimer	Club's responsibility limits regarding event participation.
Policy Updates	Note that terms may be updated periodically.

3.14 Footer

A shared component appearing at the bottom of every page across the website.

Requirement	Description
Quick Links	Shortcuts to Home, Events, About, Executives, Gallery, Contact, Privacy Policy, Terms & Conditions.
Social Media Icons	Links to the club's social media pages.
Contact Summary	Short contact info (email/phone).
Copyright Notice	"© 2026 University Computer Club. All rights reserved."
Department/University Credit	Name of the university and department, with logo (optional).
Developer Credits	Short credit section for the website development team members.

4 Governance Model (from the Club Constitution)

The internal user panel is not an independent design — it is a direct digital implementation of the Constitution (Amended 2026). The key rules the system must encode are summarized below.

4.1 Membership (Constitution Ch. 2)

Rule	Detail
Eligibility	Students of the Department of CSE, Netrokona University, or field experts (§2.1.1).
Categories	Member by power (Departmental Head), General Member, Lifetime Member (§2.1.2).
General Member	All current CSE students by default; annual fee Tk. 50.
Lifetime Member	One-time fee Tk. 5,000 via application to the President; retains all rights except voting in elections.
Cancellation	For constitutional violations, 1+ year unpaid dues, 3 consecutive unexcused absences, voluntary exit, or disciplinary/legal cause (§2.3–2.4), decided by 2/3 of the Executive Committee.
Reinstatement	Formal application to the President; 2/3 Executive Committee approval; all dues must be cleared (§2.5).
Fee changes	Only the General Assembly — not the Executive Committee alone — can revise membership fees (§2.6).

4.2 Club Structure (Constitution Ch. 3)

- General Committee — all general members; can attend and vote in the General Assembly.
- Executive Committee — elected administrators of the club; brings major decisions to the General Assembly for approval.
- 20 defined posts (§3.2), including President, Vice-President, General Secretary, Assistant General Secretary, Treasurer, Organizing Secretary (+ Assistant), Programming Secretary (+ Assistant), Event Management Secretary (+ Assistant), Editor Secretary (+ Assistant), Information Secretary (+ Assistant), Sports Secretary (+ Assistant), Publicity & Publication Secretary (+ Assistant).
- Post eligibility by academic year (§3.3): e.g., President/VP/Treasurer/Programming Secretary — final year only; General Secretary and several posts — third year; several assistant posts — second year; two posts — first year.

4.3 Roles & Responsibilities (Constitution Ch. 4, selected)

Post	Key Responsibility
Advisor	Faculty oversight; can dismantle the committee or remove a member based on General Assembly reaction.
President	Highest authority in command; signs all publishing, notices, and bills.
Vice-President	Assumes President's duties in their absence.
General Secretary	Owens publishing/arranging operations; reviews and signs all notices.
Treasurer	Manages all club finances; sole authority to issue budgets after Advisor/President approval; signatory on the bank account.
Event Management Secretary (+ Assistant)	Executes fieldwork for events alongside the Organizing Secretary.
Information Secretary (+ Assistant)	Keeps members updated; conveys club status and updates.
(Each "Assistant" post)	Assumes the responsibilities of its senior post in their absence; otherwise a proactive member.

4.4 Assembly & Elections (Constitution Ch. 5–6)

- General Assembly: 15 days' notice (general council) / 7 days (executive council). Emergency: 7/3 days. Urgent: 2 days/24 hours (§5.1).
- Quorum: 1/3 of members (§5.2).
- At least 2 General Assembly meetings per executive term are mandatory: one at the start, one just before the term ends (§5.3).
- Elections are run by the Election Committee under detailed candidate and voter rules (§6.1–6.2), including secret ballot voting and year-based post eligibility.
- The Election Commission consists of 1 Chief Election Commissioner + 4 Assistant Election Commissioners, selected by the general council, and must remain politically neutral (§6.3).

4.5 Finance & Policy (Constitution Ch. 8–9)

- Fund withdrawals require any 2 of 3 signatures: President, General Secretary, Treasurer.
- Half-yearly financial reports must be verified by the Advisors and released to all members for transparency.

- Constitutional amendments can only be added at election time; typo/interpretation fixes may be corrected anytime without changing intent.

5 User Panel — Roles & Committee Lifecycle

5.1 User Roles

#	Role	Description
1	Election Committee	Temporary committee that runs elections and forms the next Executive Committee.
2	Executive Committee	The elected body (President, General Secretary, Treasurer, and other posts per ch.3.2) that runs the club for a one-year term.

Only these two roles get a role-specific dashboard on login (Section 6.5).

No login for Advisory Board & Club Member: The Advisory Board (উপদেষ্টা মন্ডলী) and general Club Members do not get system accounts or dashboards. Advisory Board oversight happens outside the system (physical meetings, hardcopy/emailed reports); Club Members interact only with the public website — browsing events, notices, and content, and submitting the Join Us and Contact forms — without logging in. Where a governance action needs their input (e.g., a half-yearly financial report, or filling an attendance form as a meeting participant), the system exposes it as a public-facing form or a downloadable document rather than a login-gated feature.

5.2 Committee Lifecycle & Governance Workflow

This is the core lifecycle the system must enforce, repeating annually:

- The Election Committee runs an election through the Election Module and produces a complete, newly elected Executive Committee. Candidate registration only opens once the Election Commission is active; the Election Module handles nomination applications, fee verification, and symbol allotment digitally (Section 6.1.1), but the actual vote is cast offline (in person). The Election Commission records the offline outcome and enters/declares the final winners in the system.
- Once the Election Committee successfully hands over a full Executive Committee, the Election Committee is automatically dissolved — its accounts lose access and it no longer appears as active.
- The new Executive Committee's term is exactly one year.

When the term ends:

- The outgoing Executive Committee forms a new Election Commission.
- The outgoing Executive Committee's President grants the Chief Election Commissioner access to the Election Module.
- The Election Commissioner formally declares the outgoing Executive Committee dissolved, removes their access/roles, and runs a fresh election to produce the next Executive Committee.
- Once the new Executive Committee takes charge, the Election Commission is dissolved in turn.

This cycle repeats every year: Executive Committee → forms Election Commission → Election Commission elects new Executive Committee → old Executive Committee dissolved → Election Commission dissolved.

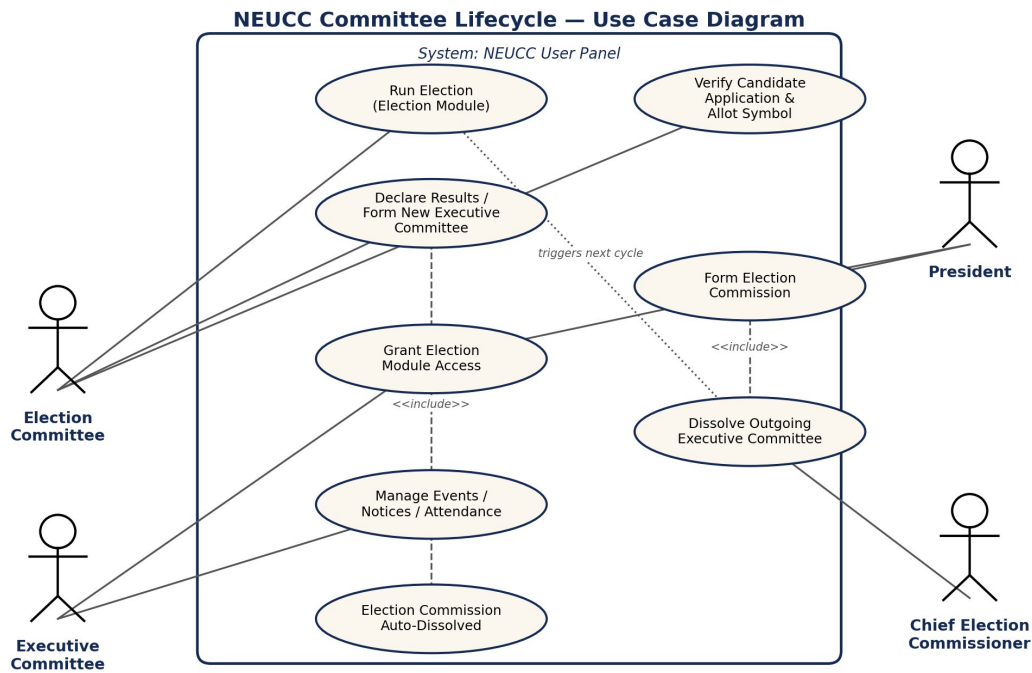


Figure 5.2 — Committee Lifecycle Use-Case Diagram: actors (Election Committee, Executive Committee, President, Chief Election Commissioner) and their governance use cases.

5.3 System Requirements for the Lifecycle

Requirement	Description
Committee status field	Every committee record (Election Committee / Executive Committee) has a status: `Active` or `Dissolved`.
Auto-dissolution trigger	Completing a handover action automatically sets the outgoing committee's status to `Dissolved` and revokes its members' elevated permissions.
Term tracking	Each Executive Committee record stores a `start_date` and `end_date` (1 year), driving the renewal workflow.
Access grant action	The Executive Committee President has a specific permission: "Grant Election Module Access", assignable only to the Chief Election Commissioner.
Dissolution action	The Chief Election Commissioner has a specific permission: "Dissolve Executive Committee", which removes the outgoing committee's roles and hands responsibilities to the newly elected members.
Audit trail	Every dissolution, handover, and access-grant action is logged with timestamp and actor (supports Constitution §9 transparency).

5.4 Role-wise Permission Matrix

Module / Action	Election Committee	Executive Committee
View dashboard	Yes	Yes
Manage Election Module	Yes (while active)	No (except President's access-grant action)
Dissolve outgoing Executive Committee	Yes (Chief Election Commissioner only, once	No

Module / Action	Election Committee	Executive Committee
	granted access)	
Grant Election Module access	No	Yes (President only)
Create / edit / delete events	No	Yes (Event Manager & Assistant Event Manager only)
View events	Yes	Yes
Publish notices / announcements	Yes (Election notices only)	Yes (Notice & Publication Team only)
View notices / announcements	Yes (General + Election only)	Yes (General + Internal)
Create monthly resolutions	No	Yes (Information Secretary & Assistant only)
Store/manage uploaded files & documents	No	Yes (Information Secretary & Assistant only)
Create & share attendance forms	No	Yes (Information Secretary & Assistant only)
Fill up attendance form	—	Yes (as meeting attendee)
View full attendance data	No	Yes full (Information Secretary & Assistant); Yes view-only (President & General Secretary)
View own submitted attendance	—	Yes
View own post's role/responsibility (from Constitution) on profile/dashboard	Yes	Yes
View full Constitution document	Yes	Yes
Manage financial records (income & expense)	No	Yes (Treasurer only)

6 User Panel — Module-wise Functional Requirements

6.1 Election Module

Requirement	Description
Election creation	Election Committee (or an authorized Chief Election Commissioner) configures an election: post list, eligible batches (per §4.2/§3.3), candidate submission/verification deadline, and the offline voting date announced in the election notice.
Candidate application	Eligible members apply for a post through an online candidate form, per eligibility rules in Constitution §6.1. Full application, fee, verification, and symbol-allotment workflow in Section 6.1.1.
Voting method (offline)	Actual voting is conducted offline (in person, on the announced date), per the voter list maintained by the Election Commission under Constitution §6.2. The system does not run digital polling or record individual votes; its role ends at symbol allotment (Section 6.1.1) and resumes when the Election

Requirement	Description
	Commission enters the final result.
Vote confidentiality	Since voting happens offline, ballot secrecy is maintained by the Election Commission through its own physical process; the system only stores and publishes the final post-wise outcome, never any individual's vote.
Result publication	Once the offline vote is counted, the Election Commission manually enters the post-wise outcome into the system and publishes the results, triggering formation of the new Executive Committee.
Result declaration download	Once results are published, every logged-in user can download the official result declaration (post-wise winners, symbols, and unopposed/walkover declarations) as a PDF, for official record-keeping or hardcopy submission.
Handover trigger	Publishing a complete result set for all posts triggers the Election Committee's dissolution (Section 5.2).
Access control	Only the active Election Committee (or a Chief Election Commissioner holding granted access) can operate this module.
Registration visibility gating	The candidate nomination/registration form (Section 6.1.1) is not shown to members at all times — it only becomes visible and submittable once the Election Commission is formed and active, and closes at the application deadline it sets.
Email notifications	Executive Committee and Election Commission members receive automated email notifications for key election milestones (election announced, application verified/rejected, symbol allotted, results published) — the only two roles with login accounts, and therefore the only ones the system emails.

6.1.1 Candidate Nomination, Fee & Symbol Allotment Workflow

Digitizes the Election Commission's candidacy process end-to-end — from the announcement notice through fee payment, verification, symbol allotment, and uncontested-post declarations.

Requirement	Description
Election announcement notice	The Election Commission publishes an Election-scope notice (Section 6.3.4) announcing the election date, the post-wise candidacy conditions (per Constitution §3.3/§6.1), and instructions for filling up the online candidate form.
Online candidate application form	After the announcement notice is published, eligible members apply for a post directly on the website. The form requires the applicant's academic year/batch, and — based on that batch — only shows the posts they are currently eligible to contest (per Constitution §3.3/§6.1); ineligible posts are not selectable. The applicant then confirms membership status and submits the required details.
Application fee	Each candidate application requires payment of an application fee, with the amount set by the Election Commission for that election.
Offline payment — bank/mobile transfer	The applicant selects this method and manually enters payment details (amount, method, transaction/reference number, date) as part of the form submission.
Offline payment — paid to Election Committee member	The applicant selects this method and simply picks the name of the Election Committee member the payment (e.g., cash) was handed to, from a list of active Election Committee members. No transaction/reference number or

Requirement	Description
	other payment detail is required for this method.
Online payment (future placeholder)	An Online Payment option is shown in the form UI but is disabled/non-functional at launch. It is reserved for a future integration and requires no working payment gateway in this phase.
Application verification	The Election Commission reviews each submitted application — eligibility and, for offline payments, the submitted payment details (transaction/reference number, or the named committee member for a cash-to-member payment) — and must explicitly verify it before the application status changes from Pending to Completed/Verified.
Symbol (প্রতীক) allotment	Once an application is verified/completed, the Election Commission allots an election symbol to the candidate for their post.
Uncontested post handling	If, after the application deadline, a post has exactly one verified candidate, no symbol is allotted for that post. The Election Commission instead declares that candidate elected unopposed (walkover), since they win without a contest; no vote is held for that post.
Eligibility fallback (open candidacy)	If, by the application deadline, a post has received no eligible (batch/year-qualified) candidate, the Election Commission reopens that specific post to all general members regardless of academic year/batch, for a short additional window it sets. Once reopened, any member may apply for that post; other posts that already have eligible candidates are unaffected.
Application status tracking	Candidates can see their own application's status — Pending, Verified/Completed, Symbol Allotted, Elected Unopposed, or Rejected — from their dashboard.
Nomination form download	Once the Election Commission verifies/finalizes an application (status Verified/Completed or later), the candidate gets a download option on their dashboard for a formatted copy of their nomination form (candidate details, post, and — once assigned — the allotted symbol) as a PDF, for their own record or printing.
Access restriction	Only the active Election Committee can verify applications, allot symbols, and declare uncontested winners. Candidates can only view and submit their own application.

6.2 Event Management Module

Requirement	Description
Create event	Event Manager and Assistant Event Manager (Executive Committee posts) create new events (name, date, description, guests, registration link, etc.), which feed directly into the public Event Tab (Section 3.2).
Edit event	Event Manager and Assistant Event Manager update event details after creation.
Delete event	Event Manager and Assistant Event Manager can delete an event.
View-only access	All other roles (Election Committee, other Executive Committee members) can only view published events on their dashboard. The general public — including Club Members and the Advisory Board, neither of whom have login accounts — views the same published events via the public Event Tab (Section 3.2), with no login required.

6.3 Notice, Resolution & Document Module

Requirement	Description
File storage	Information Secretary and Assistant Information Secretary upload and store club files/documents (e.g., meeting minutes, certificates, reports).
Notice publishing	Information Secretary, Assistant Information Secretary, Editor Secretary, Assistant Editor Secretary, Publicity & Publication Secretary, and Assistant Publicity & Publication Secretary (together, the "Notice & Publication Team") publish General/Public and Internal notices/announcements; General/Public notices additionally populate the public Announcements Page (Section 3.8). Any member of the Election Committee may separately publish Election-scope notices (see Section 6.3.4).
Monthly resolution creation	Information Secretary and Assistant Information Secretary create monthly meeting resolution records, matching the club's existing format: Meeting No., Memo No., Date, Time, Venue, President, Convener, Agenda, Discussion, Resolution/Decision, Implementation Responsibility table, Attendee Count, and Signatories.
Notice & resolution download	Any published notice and any monthly resolution record can be downloaded as a PDF for offline/hardcopy use, directly from wherever it is listed (Announcements Page, dashboard notice feed, or resolution history). Download access always follows the same visibility/permission rules already governing that record (Section 6.3.4) — e.g., a Club Member cannot download an Internal or Election-scope notice they are not permitted to view.
Access restriction	File storage and monthly resolution records can only be created/edited by Information Secretary and Assistant Information Secretary. General/Public and Internal notice content creation and publishing is open to the full Notice & Publication Team; Election-scope notice creation and publishing is open to any Election Committee member (see Section 6.3.4). Other roles have read access based on their permission level and the notice's visibility scope.

6.3.1 Notice Pad (Letterhead) Template

The official NEUCC notice pad is a fixed, bilingual letterhead design, based on the club's existing paper format. Every generated notice must reproduce this layout exactly.

Zone	Content
Header — left	Club name, department, and university name in Bangla ("কম্পিউটার ক্লাব", "সিএসই বিভাগ", "নেত্রকোণা বিশ্ববিদ্যালয়", "নেত্রকোণা-২৪০০")
Header — center	University crest/logo + NEUCC club logo, side by side
Header — right	Club/department/university name in English ("Computer Club", "CSE Department", "Netrokona University", "Netrokona-2400")
Header divider	A horizontal rule separating the header block from the body
Reference line	"তারিখ:" (Date) on the left and "স্মারক নং:" (Memo No.) on the right, on the same line
Subject line	"বিষয়:" (Subject) — one-line summary of the notice
Body	Free-form notice text (paragraphs, and where relevant a structured list such as convener/member-secretary/members)

Zone	Content
Signature block	Right-aligned: authorization line (e.g., "সভাপতির অনুমতিক্রমে"), signature image/space, date, signatory name, designation, and "কম্পিউটার ক্লাব"
Distribution list ("অনুলিপি")	Numbered list of who the notice/copy is sent to (e.g., President, General Secretary, relevant committee, club file)
Watermark	Faint centered "NETROKONA UNIVERSITY COMPUTER CLUB" logo/wordmark behind the body text

6.3.2 Notice Pad Generator — Workflow

A dedicated sub-feature for producing official, letterhead-formatted notices using the template in 6.3.1. Restricted to the Notice & Publication Team (Information Secretary, Assistant Information Secretary, Editor Secretary, Assistant Editor Secretary, Publicity & Publication Secretary, and Assistant Publicity & Publication Secretary).

Step	Requirement	Description
1	Official pad/letterhead template	The system renders every notice using the fixed pad layout defined in Section 6.3.1.
2	AI-assisted content drafting	Any member of the Notice & Publication Team can invoke the Agentic AI Content Generator (Section 6.3.3) to draft the subject, body, and structured fields of the notice.
3	Manual editing	The AI-drafted (or manually written) content is fully editable before finalizing — text, formatting, and details can be changed freely.
4	Memo No. & Date entry	A member of the Notice & Publication Team manually enters the স্মারক নং (Memo No.) and তারিখ (Date) for the notice; these are not AI-generated, since they follow the club's official numbering sequence.
5	"Generate on Pad" action	A dedicated button places the finalized content — including the entered Memo No. and Date — onto the official notice pad/letterhead layout, producing a clean, formatted notice.
6	Output — Image	Generates a rendered image file (e.g., PNG/JPG) of the finished, letterhead-formatted notice.
7	Output — PDF	Simultaneously generates a PDF version of the same finished notice.
8	Storage	The generated PDF is saved/archived in the system's document storage for record-keeping (linked to the Notice/Resolution history, indexed by Memo No.).
9	No auto-publish	Generating the pad image/PDF does not automatically publish the notice anywhere (e.g., not auto-posted to the public Announcements Page). Publishing remains a separate, explicit action by a member of the Notice & Publication Team.
10	Access restriction	Only the Notice & Publication Team (Information Secretary, Assistant Information Secretary, Editor Secretary, Assistant Editor Secretary, Publicity & Publication Secretary, and Assistant Publicity & Publication Secretary) can access the AI drafting tool and the "Generate on Pad" action.

6.3.3 Agentic AI Content Generator

Rather than a one-shot "write this text" tool, the notice-drafting assistant behaves as an agent: it gathers context from the system, drafts a complete structured notice, checks its own output, and asks for missing information instead of guessing.

Requirement	Description
Brief-to-draft generation	A member of the Notice & Publication Team gives a short instruction (e.g., "Notice for forming a 3-member constitution review committee") and the agent produces a complete draft — subject line, formal body text, and any structured fields the notice type needs.
Context retrieval from club records	The agent pulls relevant data already in the system — current Executive Committee members and designations, past similar notices, active committees, member names/roles — to auto-fill names, designations, and references instead of asking the secretary to retype them.
Structured field population	For notices with defined sub-structure (e.g., a committee-formation notice: আহ্বায়ক/Convener, সদস্য সচিব/Member-Secretary, সদস্যবৃন্দ/Members, submission deadline), the agent proposes each field separately so it can be reviewed and edited field-by-field, not just as one paragraph.
Distribution list ("অনুলিপি") suggestion	Based on the notice type, the agent suggests a standard recipient/copy list (e.g., President, General Secretary, relevant committee, club file) drawn from the roles defined in the Constitution and permission matrix (Section 5.4), for the secretary to confirm or edit.
Tone & format consistency	The agent matches the formal Bangla/English tone and structure of the club's existing notices (as established in prior Notice/Resolution records) so new notices read consistently with past ones.
Clarifying questions over guessing	If required information is missing or ambiguous (e.g., no deadline given, unclear committee size), the agent asks the secretary a direct question rather than inventing a plausible-sounding value.
Iterative refinement	The secretary can give follow-up instructions ("make it more formal," "add these two names," "shorten the second paragraph") and the agent revises the existing draft in place rather than starting over.
Self-check before handoff	Before handing the draft to the secretary for final review, the agent checks that all required sections (subject, body, any structured list, and — where applicable — a deadline and distribution list) are present, and flags any that are missing.
Human-in-the-loop finalization	The agent never finalizes or generates the pad image/PDF itself — every draft must be explicitly reviewed and approved by a member of the Notice & Publication Team before the "Generate on Pad" action (Section 6.3.2, Step 5) is used.
Memo No. & Date excluded from AI scope	The agent does not generate or guess the Memo No. or Date (Section 6.3.2, Step 4) — these remain a manual entry by the secretary, since they follow the club's official sequential numbering.
Access restriction	Only the Notice & Publication Team (Information Secretary, Assistant Information Secretary, Editor Secretary, Assistant Editor Secretary, Publicity & Publication Secretary, and Assistant Publicity & Publication Secretary) can invoke the agent.

6.3.4 Notice Visibility & Audience Scope

Not every notice is meant for the same audience. The system enforces three notice visibility scopes, and grants the Election Committee independent publishing rights limited to its own announcements.

Requirement	Description
Notice scope tagging	Every notice created in the system is tagged with a visibility scope at creation time: General/Public, Internal, or Election.
General/Public scope	Visible to everyone — appears on the public Announcements Page (Section 3.8) with no login required, and is also visible to the two logged-in roles (Election Committee, Executive Committee) on their dashboards.
Internal scope	Visible only to Executive Committee. Hidden from Election Committee and never appears on the public Announcements Page. Since the Advisory Board has no login, its oversight of internal notices happens outside the system (e.g., shared as hardcopy/email at physical meetings), not through the notice module.
Election scope	Visible only to Election Committee. Hidden from Executive Committee, and never appears on the public Announcements Page.
Who can publish which scope	The Notice & Publication Team can create/publish General/Public and Internal notices only. Any Election Committee member can create/publish Election-scope notices only, using the same notice pad workflow (Section 6.3.2) with manual drafting; the Agentic AI Content Generator (Section 6.3.3) remains exclusive to the Notice & Publication Team.
Access restriction	Election Committee members cannot create/publish General or Internal notices, and the Notice & Publication Team cannot create/publish Election-scope notices. When the Election Committee is dissolved (Section 5.2), its members immediately lose access to Election-scope notice creation along with all other elevated permissions.

6.4 Attendance Module

Requirement	Description
Attendance form creation	Information Secretary and Assistant Information Secretary create an attendance form for any club meeting and share it with meeting participants.
Form submission	Executive Committee members fill up the form via their dashboard (login required). Club Members and Advisory Board members, who have no login, fill it via a shareable public link instead — entering their name, student ID, and batch manually, with no account needed.
Data storage & privacy	All submitted attendance data is stored and remains private to the Information Secretary and Assistant Information Secretary by default.
Restricted visibility	The President and General Secretary may view attendance records, but cannot edit, export, or manage them.
No other access	No other role (Election Committee, other Executive Committee posts) and no member of the public can view another person's attendance submission.

6.5 Dashboard

Requirement	Description
Universal dashboard	Every logged-in user, regardless of role, has a personalized dashboard on login.
Role-based dashboard	Dashboard widgets are filtered by role and permission — e.g., Election

Requirement	Description
content	Committee sees election status and candidate applications to review; Executive Committee posts see their module shortcuts (events, notices, attendance, treasury, etc., per Section 5.4). Advisory Board and Club Members have no dashboard, since they have no login accounts.
Committee status visibility	Dashboard indicates current committee status (e.g., "Executive Committee — Active, X days remaining in term" or "Election in progress") to keep the lifecycle transparent to relevant roles.
Post-based responsibility display	For every member of the Election Committee or Executive Committee, the profile/dashboard shows that member's specific post (e.g., President, Treasurer, Chief Election Commissioner) along with the corresponding role description and responsibilities as defined in Constitution Chapter 4 (Role of Every Post) / Chapter 6 (Election Commission responsibilities, §6.3). This is pulled directly from the constitution text, not re-written per user.
Constitution viewer	Election Committee and Executive Committee members can open and read the full Constitution document (all chapters) from within their dashboard, with their own post's specific section highlighted, and can also download the full Constitution as a PDF for offline/hardcopy reference. Since the Advisory Board and Club Members have no login, the same downloadable Constitution PDF is also published on the public website (About page, Section 3.3), so everyone can access it without an account.
Access restriction	The post-based responsibility display is available only to Election Committee and Executive Committee members (since only they hold constitutional posts); the Constitution viewer itself is available to every logged-in role.

6.6 Treasury / Finance Module

Digitizes the Treasurer's bookkeeping duty under Constitution §8. The Treasurer maintains two linked ledgers — money coming into the club and money going out — each entry backed by a supporting document.

Requirement	Description
Income ledger	Treasurer logs every amount coming into the club: the fund head/category it falls under (e.g., membership fee, sponsorship, event income, donation), the amount, the source, and the specific member/donor/organization who is giving it.
Expense ledger	Treasurer logs every amount spent by the club: the fund head/category it falls under (e.g., event cost, printing, refreshments, logistics), the amount, the purpose, and the specific member who is spending/authorizing it.
Supporting document attachment	Each income and expense entry supports an attached document (money-receipt, invoice, voucher, or bill scan/photo) as proof, stored alongside the entry for verification and audit.
Fund head / category management	Treasurer can select from (and, where needed, add) standard income and expense heads so every entry stays categorized consistently for reporting.
Running balance	The system automatically calculates and displays the club's current balance in real time from all logged income and expense entries.
Half-yearly financial report	Income/expense records can be compiled into the half-yearly financial report required by the Constitution (§8). Since the Advisory Board has no login, this report is delivered to them as a downloadable/emailed PDF for their verification, rather than through in-system access; the President reviews it within the system, and the verified report is then released to all members.

Requirement	Description
Transaction & report download	The full income/expense ledger — or a filtered subset (date range, fund head, or a single entry with its attached supporting document) — can be downloaded as PDF/Excel by the Treasurer for offline bookkeeping and audit. The compiled half-yearly financial report is likewise downloadable as a PDF once the President has reviewed it, so it can be sent to the Advisors (who have no login) and distributed to members needing a hard copy. Download availability follows the same view rights as the record itself (Treasurer: full; President: view-and-download for oversight; other roles: no access).
Access restriction	Only the Treasurer can create/edit income and expense entries. The President has view access for oversight/verification per Constitution §8; other logged-in roles do not have access to raw financial records. The Advisory Board, having no login, exercises its oversight via the downloadable half-yearly report rather than direct system access.

7 Non-Functional Requirements

Requirement	Description
Responsiveness	The website and panel must work properly on desktop, tablet, and mobile devices.
Performance	Pages should load quickly, with optimized images and minimal unnecessary scripts.
Usability	Navigation should be simple and intuitive for all types of visitors and internal users.
Security	Contact/membership form inputs validated and protected from spam/injection attacks; authentication secured via JWT; role-based access control (RBAC) enforced at the API/backend level, not just hidden in the UI.
Auditability	All committee transitions, dissolutions, notice publications, and attendance-record access are logged with timestamp and actor.
Data Privacy	Attendance data and personally identifiable information are restricted per Section 6.4 and handled per the club's Privacy Policy (Section 3.12).
Session/term awareness	The system tracks committee term dates so permissions automatically expire when a committee is dissolved.
Maintainability	Code should be well-structured (TypeScript, modular components) so future committee members can easily update content.
Accessibility	Follow basic accessibility practices (readable fonts, proper color contrast, alt text for images).
Cross-Browser Support	The site should render correctly on Chrome, Firefox, and Edge.
Dark/Light Theme Toggle	Users can switch between dark and light themes; defaults to system preference.
Notifications	Automated email notifications are sent to Executive Committee and Election Commission members for key governance actions (Section 6.1). The Advisory Board and Club Members, having no login accounts, do not receive system notifications.

8 Technology Stack

Frontend

Requirement	Description
Framework	Next.js
Language	TypeScript
Styling	Tailwind CSS
Charts/Analytics	Recharts (for stats and analytics visuals)
Authentication (Client)	Better Auth (for user login and registration)

Backend

Requirement	Description
Framework	Express.js
Language	TypeScript (mandatory)
Database	PostgreSQL
ORM	Prisma (schema, migrations, and type-safe queries against PostgreSQL)
Authentication (Server)	JWT-based authentication

Other Tools

Requirement	Description
Version Control	Git and GitHub for source code management
Hosting	Vercel (frontend) / Render or university hosting (backend)
Payment Gateway	Not implemented at launch. Only Offline Payment (manual entry, Election Commission-verified) is active for candidate application fees (Section 6.1.1); an Online Payment option is shown in the UI as a placeholder for a future gateway integration.

9 Site Navigation Flow

The overall page flow of the public website:

- Home → Event Tab (list & details in one page)
- Home → About Page → Executives Page
- Home → Contests Page / Gallery Page / Achievements Page / Announcements Page
- Home → Sponsors Page / Join Us Page
- Home → Contact Page
- Footer (visible on all pages) → Privacy Policy / Terms & Conditions

The overall flow of the internal panel (post-login):

- Login → Role-based Dashboard (Section 6.5)
- Executive Committee → Event Module / Notice & Resolution Module / Attendance Module (per post permissions)

- Election Committee → Election Module
- Election Committee & Executive Committee (the only login roles) → View-only access to events, notices, and (where permitted) attendance/committee status on their dashboard. Advisory Board and Club Members have no login and use the public website only.

10 Reference Documents

This requirement is derived from and must remain consistent with:

- The Constitution Policy of Computer Club of Netrokona University (Amended 2026) — particularly Chapters 2 (Membership), 3 (Club Structure), 4 (Role of Every Post), 5 (Assembly), 6 (Election Process), 8 (Finance), and 9 (Conclusion/Policy Betterment).
- NEUCC Monthly Meeting Resolutions (July 2025 – July 2026), which establish the working format for meeting minutes, resolutions, and committee reporting that the Notice/Resolution Module (Section 6.3) replicates digitally.

11 Conclusion

This document defines the complete requirement for the NEUCC Web Application: a fourteen-page public website that represents the club to the outside world, and a role-based internal panel — Advisory Board, Election Committee, Executive Committee, and Club Member — that digitizes the club's constitutional governance, including the annual committee lifecycle, election process, event management, notice/resolution publishing, and privacy-controlled meeting attendance. Built as a production-ready full-stack application using Next.js, TypeScript, Tailwind CSS, Express.js, PostgreSQL, and Prisma, this system will give NEUCC both a professional public identity and an accountable, constitution-compliant way to run the club digitally.